

# SHUVAM PAL

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## Education

**Dayananda Sagar College of Engineering**

**Bengaluru, India**

*B.E. Artificial Intelligence and Machine Learning*

## Experience

**Bentham AI**

*Dec 2025 – Feb 2026*

*Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions*

*Remote*

- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, filing descriptions, and operator-ready drafts with validation before handoff.
- Built a resilient automation backend with retries, session recovery, public-site API lookups, and state validation, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

**Google Summer of Code**

*May 2025 – Sept 2025*

*Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX*

*Remote*

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output regressions across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

## Open Source

[Contributions](#)

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

## Projects

**Swish Support System** | *Python, FastAPI, LLM APIs, PostgreSQL, Redis, Langfuse, Pytest*

[GitHub](#)

- Built an **AI support backend** where LLM assessments classify complaint intent, severity, ambiguity, requested resolution, and visual-evidence usefulness while deterministic services control order, refund, and escalation policy.
- Added an **evaluation and feedback loop** with response-quality checks, copy-contract guardrails, LLM-judge scoring, Langfuse traces, Redis session context, and regression tests for multi-turn support cases.

**Penny Lane Capital** | *Python, Agno, Pydantic, SQLite, Embeddings, Vector Memory, JSONL*

[GitHub](#)

- Built a **multi-agent research workflow** with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and JSONL traces.
- Implemented **vector memory and semantic lesson retrieval** with local embeddings, reward-scored loops, and validation/backtest harnesses so agent behavior and retrieval quality could be inspected across runs.

**Omni-SSM Voice Agent** | *Python, PyTorch, ONNX Runtime, Sherpa-ONNX, Moonshine, Hindi ASR, Qwen TTS, Piper*

- Built a local **voice-agent research loop** combining SSAMBA streaming CTC, Piper low-latency TTS, optional Qwen3-TTS expressive output, and WebRTC loopback benchmarks.
- Worked on **Moonshine Hindi ASR tuning** with manifest-driven dataset loading, teacher/student distillation, partial fine-tuning scripts, and WER, first-audio latency, memory, and provider-profile checks.

## Technical Skills

**Languages:** Python, Go, TypeScript, C++

**Frameworks:** FastAPI, PyTorch, ONNX Runtime, Sherpa-ONNX, Agno, Node.js, Pytest

**Tools:** PostgreSQL, SQLite, Redis, RabbitMQ, Docker, GitHub Actions, GCP, Langfuse, Prometheus, OpenTelemetry, Colab, Vector Memory